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SET : 14.

Dataset: Survey Results

Respondent Question 1 Question 2 Question 3

1 A B C

2 B A D

3 C A B

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.
2. Generate a radar chart to visualize the overall pattern of responses across all three questions.
3. Build a table to show the survey response data for each respondent. Label the table elements.
4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

SOLUTION :

CODE :

01.

```
%%R
```

```
library(ggplot2)
```

```
df14 <- data.frame(  
  Respondent = factor(c(1, 2, 3)),  
  Question_1 = c("A", "B", "C"),  
  Question_2 = c("B", "A", "A"),  
  Question_3 = c("C", "D", "B")  
)
```

```
ggplot(df14, aes(x = "Question 1", fill = Question_1)) +  
  geom_bar(width = 0.5) +  
  geom_text(  
    stat = "count",  
    aes(label = after_stat(paste0("Option ", fill, " (", count, ")")),  
    position = position_stack(vjust = 0.5),  
    color = "white",
```

```

    fontface = "bold"
  ) +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Survey Item",
    y = "Count of Responses",
    fill = "Response Choice"
  ) +
  theme_minimal() +
  scale_fill_brewer(palette = "Set1")
02. %%%R

if (!requireNamespace("fmsb", quietly = TRUE)) install.packages("fmsb", quiet
= TRUE)
library(fmsb)

radar_df <- data.frame(
  Question_1 = c(1, 2, 3),
  Question_2 = c(2, 1, 1),
  Question_3 = c(3, 4, 2)

radar_data <- rbind(
  c(4, 4, 4),
  c(1, 1, 1),
  radar_df
)
rownames(radar_data) <- c("Max", "Min", "Respondent 1", "Respondent 2",
"Respondent 3")

colors_border <- c("#E41A1C", "#377EB8", "#4DAF4A")
colors_in <- c(rgb(0.9, 0.1, 0.1, 0.2), rgb(0.2, 0.5, 0.7, 0.2), rgb(0.3, 0.7, 0.3,
0.2))

radarchart(
  radar_data,
  axistype = 1,
  pcol = colors_border,
  pfc col = colors_in,
  plwd = 2,
  plty = 1,
  cglcol = "grey",
  cglty = 1,
  axislabcol = "grey",
  caxislabels = c("A", "B", "C", "D"),
  cglwd = 0.8,
  vl cex = 1.1,
  title = "Radar Chart: Pattern of Responses Across Questions"

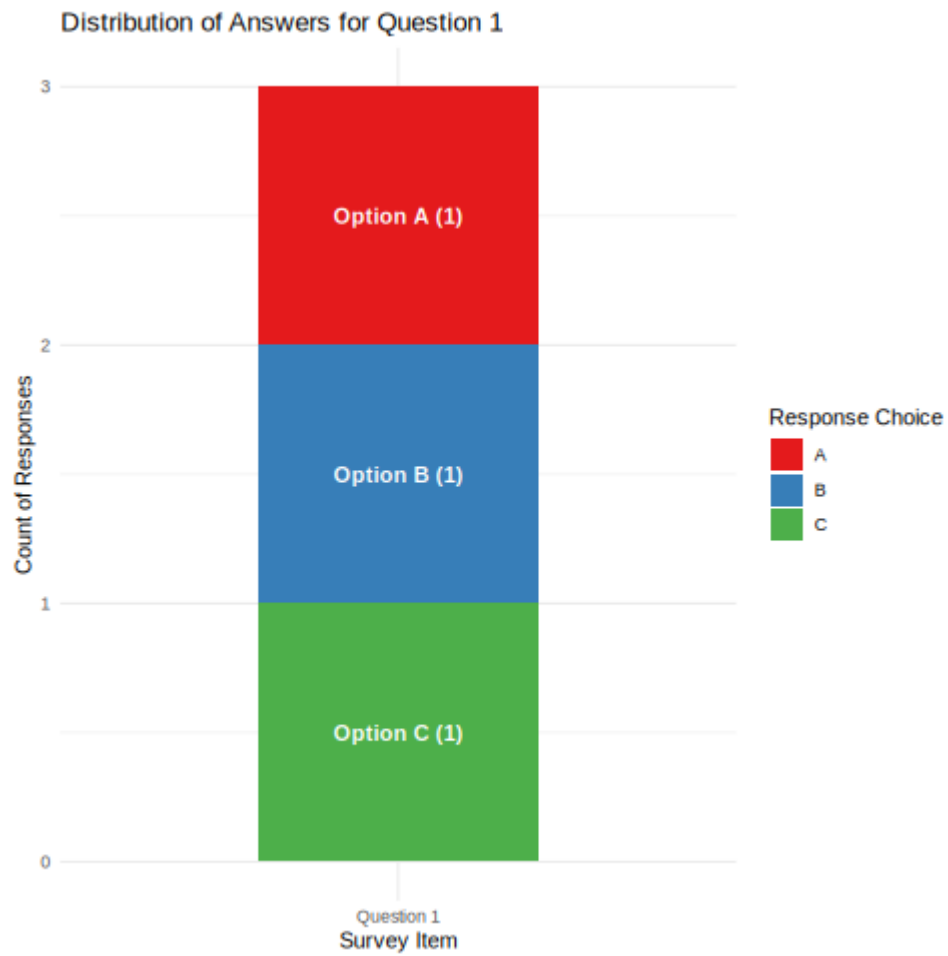
```

```
)  
  
legend(  
  x = "topright",  
  legend = c("Respondent 1", "Respondent 2", "Respondent 3"),  
  bty = "n",  
  pch = 20,  
  col = colors_border,  
  text.col = "black",  
  pt.cex = 2  
)  
03 %%R
```

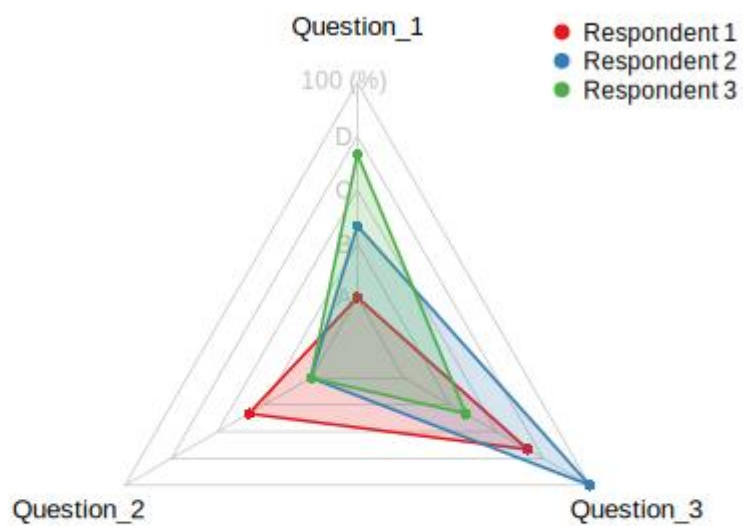
```
df14_table <- data.frame(  
  Respondent_ID = c(1, 2, 3),  
  Question_1 = c("A", "B", "C"),  
  Question_2 = c("B", "A", "A"),  
  Question_3 = c("C", "D", "B")  
)
```

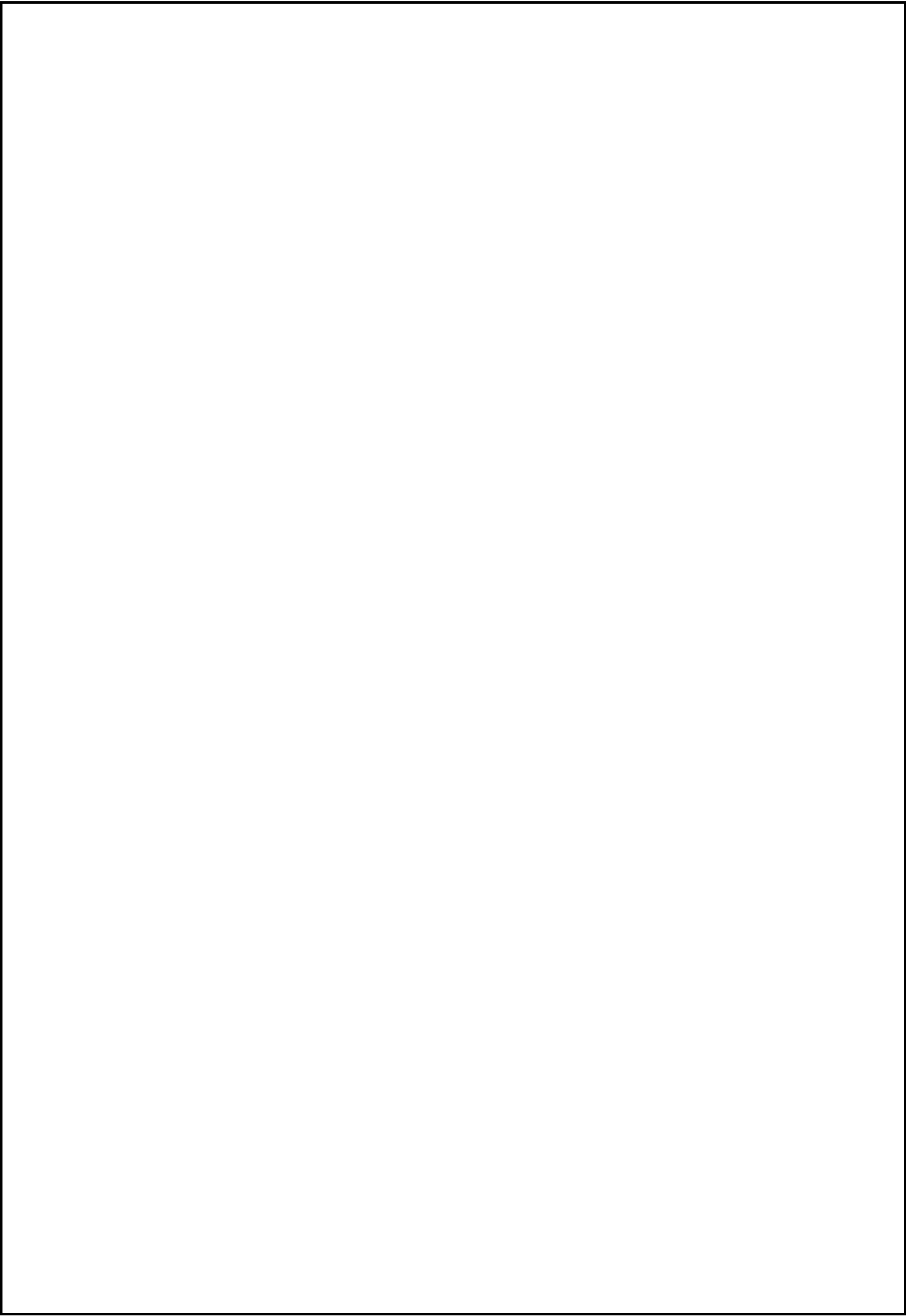
```
print("--- SURVEY RESPONSE DATA TABLE ---")  
print(df14_table)
```

OUTPUT :



Radar Chart: Pattern of Responses Across Questions





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